Points of Inflection

Exam code: 9MA0

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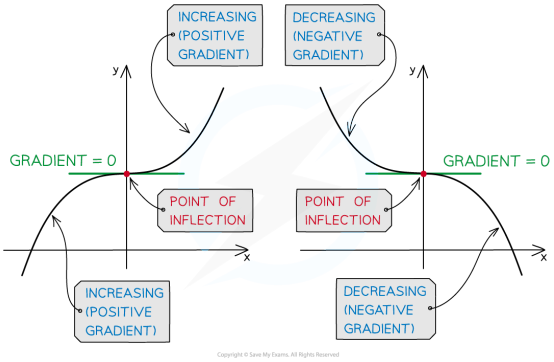
Points of inflection

What is a point of inflection?

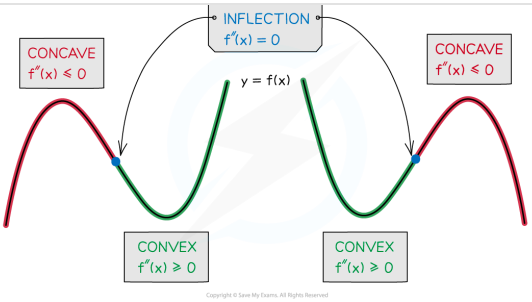
- At AS level you encountered points of inflection when discussing **stationary points**
  - When the sign of the first derivative (ie of the gradient) is the same on both sides of a stationary point, then the stationary point is a point of inflection



- A point of inflection does not have to be a stationary point however
- A **point of inflection** is any point at which a curve changes from being convex to being concave
  - This means that a **point of inflection** is a point where the **second derivative changes sign** (from positive to

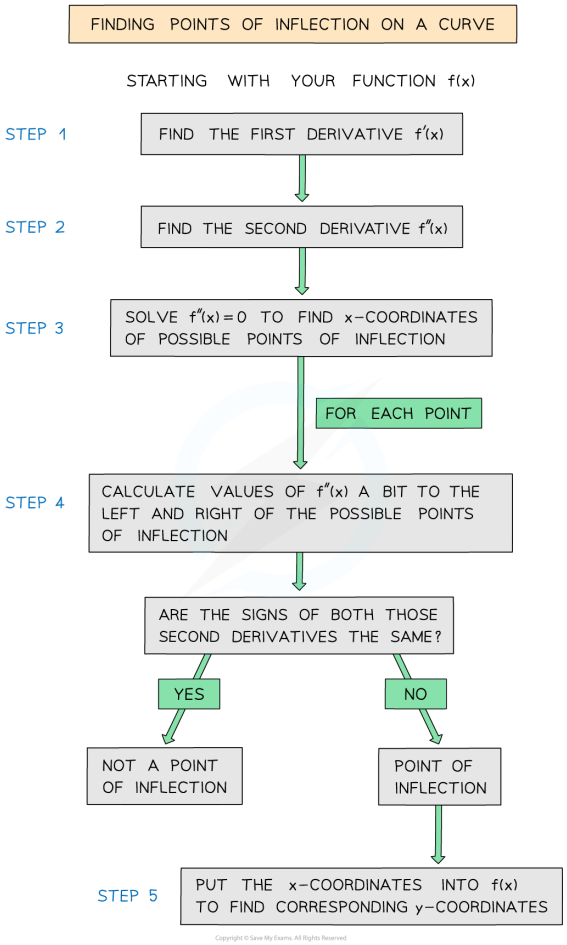
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



How do I find points of inflection?

- To find the points of inflection of a curve with equation $y = f(x)$:



(i.e. A LITTLE BIT MORE THAN) THE x COORDINATE OF A POSSIBLE POINT OF INFLECTION

- **DO** CHOOSE A CONVENIENT POINT WHERE POSSIBLE ($x = 0$ AND $x = 1$ ARE BOTH VERY EASY TO PUT INTO FORMULAS FOR EXAMPLE)
- **DO NOT** HOWEVER CHOOSE AN x VALUE TO THE LEFT OR RIGHT THAT JUMPS PAST THE x–COORDINATE OF ANOTHER POSSIBLE POINT OF INFLECTION!

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Examiner Tips and Tricks

- Remember – the first derivative (ie the gradient) does NOT have to be zero at a point of inflection!

Worked Example

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